B. Tech. Fifth Semester (Information Technology)/(SOE_23 IT-101) Examination

Course Code : 23 IT 1504 **Course Name : Machine Learning**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.
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(5) Assume suitable data wherever necessary and mention at the beginning of answer.
(6) Illustrate your answers wherever necessary with the help of neat sketches.

1. (a) (i) Explain Logistic regression using formula for predicting the probability of a class.

(ii) Let the coefficients of logistic regression are found to be :

$\beta_0 = -8.9035$, $\beta_1 = 0.0830$, $\beta_2 = 0.1981$, $\beta_3 = 0.3637$

where :

- β_0 = intercept,
- β_1 = coefficient of **Age**,
- β_2 = coefficient of **BMI**,
- β_3 = coefficient of **Gender** (1 = Male , 0 = Female).

Write the equation for the logistic regression model and find whether an individual with **Age = 55 years , BMI = 28.5** and **Gender = Male** is likely to have hypertension. 8(CO1)

OR

(b) Given two major differences between parameters and hyperparameters of ML models.

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Student's salary prediction model is trained and has the equation –
$y' = 0.62x1 + 0.41x2 - 1.13$
where vector $x = [x1, x2] = [CGPA, CommSkill]$ and y' is predicted salary in lakh. Then answer the following :

(i) Is it a regression or classification Model ?

(ii) What are the predictors and target Variable ?

(iii) Is feature scaling required here if all the features are in the range [0, 10] ?

(iv) If a student has $x = [7.1, 6.5]$, then what is his predicted salary ?

(v) Can addition of gender feature improve the prediction of salary ? Justify.

(vi) Suggest any two ML algorithms to solve this problem. 8(CO1)

2. (a) (i) Give the algorithmic steps for the KNN classification algorithm.

(ii) Apply KNN algorithm on following dataset to find the value of y when $x_1 = 4$ and $x_2 = 6$ for the values of $K = 3$ and $K = 2$. Which of K is better and why ? (Note : Do not normalize the data.) Use Euclidean distance. Is this a linearly separable pattern space ? Justify by plotting the feature vectors in 2 - D space.
Dataset :

x_1	x_2	y
1	6	Yes
4	5	No
6	4	No
2	6	Yes
8	4	Yes
9	10	No
10	3	Yes
8	8	No
7	2	Yes

$\sqrt{(x_1-4)^2+(x_2-6)^2}$

9(CO2)

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(b) (i) Explain Naïve Bayes Classification approach in brief.

(ii) Consider COVID-19 detection model trained with Naïve Bayes' approach using the following data. Find the probability that the patient X new with symptoms as (Fever = YES, Cough = YES and Breathing Issue = NO) is COVID positive.

ID	Fever	Cough	Breathing Issues	Infected
1	NO	NO	NO	NO
2	YES	YES	YES	YES
3	YES	YES	NO	NO
4	YES	NO	YES	YES
5	YES	YES	YES	YES
6	NO	YES	NO	NO
7	YES	NO	YES	YES
8	YES	NO	YES	YES
9	NO	YES	YES	YES
10	YES	YES	NO	YES
11	NO	YES	NO	NO
12	NO	YES	YES	NO
13	NO	YES	YES	NO
14	YES	YES	NO	NO

9(CO2)

3. (a) (i) Suppose we have three classes : Class 1, Class 2 and Class 3. Each class has a One-vs-All (OvA) SVM classifier represented by a weight vectors w and bias b.
Suppose the weight vectors and biases for the classifiers are :

Classifier for (Class 1 vs. Class 2 & Class 3) : $w1 = (2, 1), b1 = -4$

Classifier for (Class 2 vs. Class 1 & Class 3) : $w2 = (-1, 2), b2 = 1$

Classifier for (Class 3 vs. Class 1 & Class 2) : $w3 = (1, -1), b3 = 0$

Then classify the test point (2, 3).

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(ii) How would you do One vs. One (OvA) classification of test instance if –

OvO (1 vs. 2) classifier predicts it as Class 1,

OvO (1 vs. 3) classifier predicts it as Class 1 and

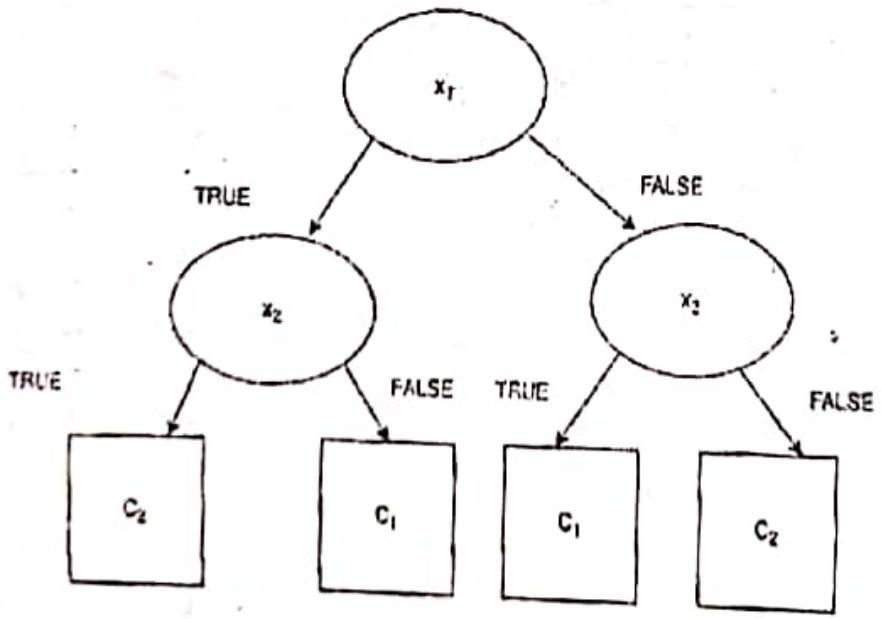
OvO (2 vs. 3) classifier predicts it as Class 2 ? 9(CO2)

OR

(b) (i) Write and explain any two hyperparameters of a decision tree.

(ii) Apply the knowledge of working of a decision tree to complete the dataset from which the given decision tree is derived (the format for dataset is given as table below). Here predictors x_1 and x_2 are Boolean any y is also Boolean. Consider C_1 as YES and C_2 as NO. The behaviour of y is similar to which logic gate.

x_1	x_2	y (C_1 or C_2)
T	T	C_2
T	F	C_1
F	T	C_1
F	F	C_2



9(CO2)

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4. (a) (i) Give a pseudocode for k-means clustering algorithm.

(ii) A travel recommendation company has collected data about 200 popular tourist locations in India. Each location is described using the following features :

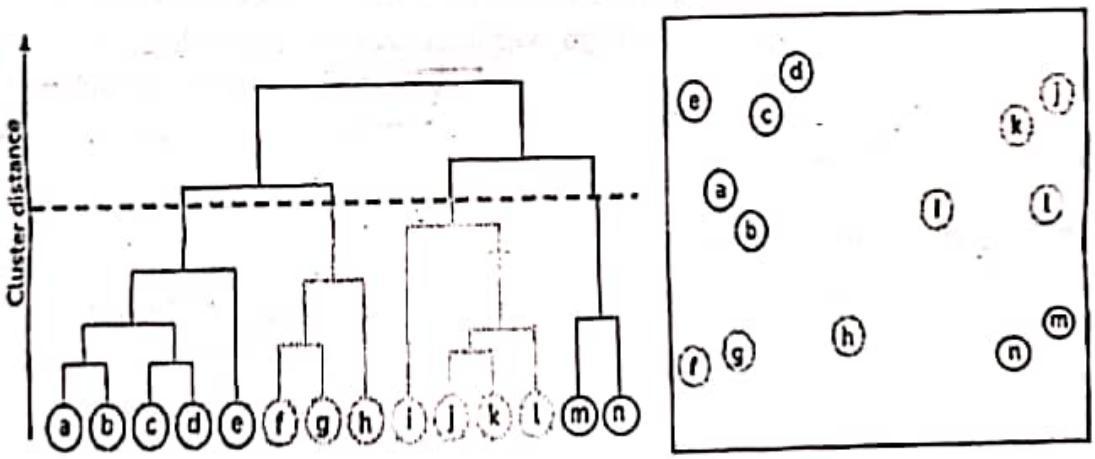
Average cost per day (in ₹), number of visitors per day, Type of region (Categorical : Hill Station, Coastal, Historical, Urban, Wildlife). The company wants to group similar tourist destinations to Offer customized travel packages and to Optimize promotion strategies for different traveller profiles (budget, adventure, cultural interest).

Analyze the problem and suggest an algorithm to achieve this task and justify your choice. What is the use of these groups formed ? 8(CO3)

OR

(b) (i) What is dendrogram ? Explain its role in hierarchical clustering.

(ii) Analyze the following dendrogram and answer questions given below :



A sliced dendrogram with its resulting clusters mapped to its corresponding feature vector space on the right side

(i) How many clusters are formed by the cluster distance indicated by the dotted line ?

(ii) Draw another horizontal dotted line at suitable height that gives 7 clusters. 8(CO3)

Contd.

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5. (a) (i) Explain the role of confusion matrix in classification.

(ii) Find the accuracy and F1-Score for the following confusion matrix for RTPCR test. Assume that 5000 tests were administered on people. The Confusion matrix is :

Actual \ Predicted	+ ve	− ve
+ ve	TP = 27	FN = 19
− ve	FP = 04	TN = 4950

Which type of error is more crucial in this case (Type-I or Type-II) and why ? 8(CO4)

OR

(b) (i) Explain the process of k-fold cross validation of ML model.

(ii) You are comparing the performance of two machine learning models using k-fold cross-validation. Model A achieves a higher mean accuracy but with high variance across the folds, while Model B has slightly lower mean accuracy but much lower variance. Evaluate which model is more reliable for deployment and justify your reasoning. 8(CO4)

6. (a) (i) What is CNN ?

(ii) Apply Max pooling on the given 4 by 4 image with dropout mask of size 2 by 2 given below and stride equal to 2. Report the new image.

9	2	6	5
2	1	6	6
7	5	4	6
8	9	2	5

Image

1	0
1	1

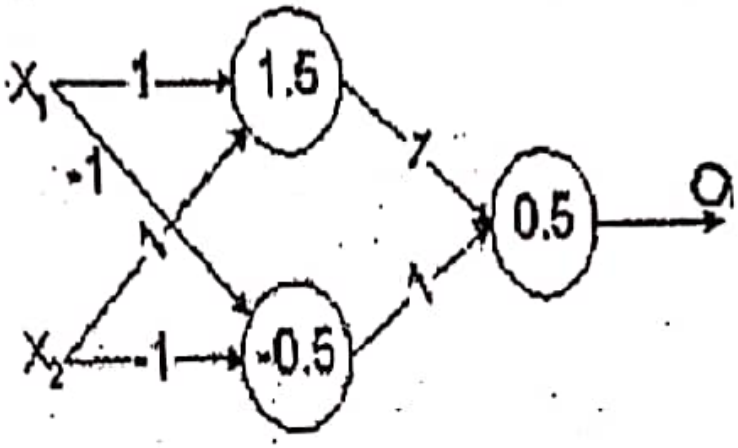
Dropout mask 8(CO2)

OR

(b) (i) What is MP Neuron ? Give its structure.

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(ii) Consider the following ANN made from MP Neurons and find the output O of the network for all possible combinations of two binary inputs X_1 and X_2. Which logical gate is implemented by this network ?



8(CO2)

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(7) Use of non - programmable calculator is permitted.

1. Solve any **One** :

(a) Define Machine Learning and explain its types. Also, Differentiate between classification and regression using appropriate examples. 8(CO1)

(b) Explain the following terms :

- Data sets (Training, Testing, Validation).
- Hyper parameters.
- Epoch.
- Data quality on performance of ML algorithm.
- Feature selection.
- Ethics in doing ML experiments.
- Feature scaling. 8(CO1)

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2. Solve any **One** :

(a) Consider the dataset of pre-owned cars containing car age and its sale price. Using this data predict the sale price of a car with the age of 7.5 years using :

(a) Linear regression with model parameters $\beta_0 = 7.51$, $\beta_1 = -0.55$.

(b) KNN with $k = 2$.

CarAge = [1.5 , 2 , 3 , 3.2 , 4 , 4.5 , 5.3 , 6 , 6.5 , 7 , 8.2]

SalePrice = [7 , 6.1 , 6.2 , 5.5 , 5.2 , 5.3 , 4.2 , 4.3 , 4 , 3.9 , 3] 9(CO2)

(b) Write the pseudocode for the KNN regression algorithm.
It is required to predict the sale price of used car based upon the age of a car and the mileage. The training data for the model is given below.

Car age	Mileage	Sale Price
2	11200	3.55
5	24500	3.1
2	11500	3.4
2	25600	3.25
4	29999	2.9
6	44500	2.5
5	19500	3.4
1	16500	3.95
4	32100	3.5
3	19904	3.65
7	66570	2.1

Use KNN Regression model (with $k = 2$) to predict the sale price of used car with of age = 3 years and mileage = 16500 km. Apply **feature normalization** (Min-Max scaling) on the input features. 9(CO2)

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3. Solve any **One** :

(a) What are Decision Trees ? Write and explain the following terms in brief :

(1) Entropy.

(2) Information Gain.

(3) Gini Index.

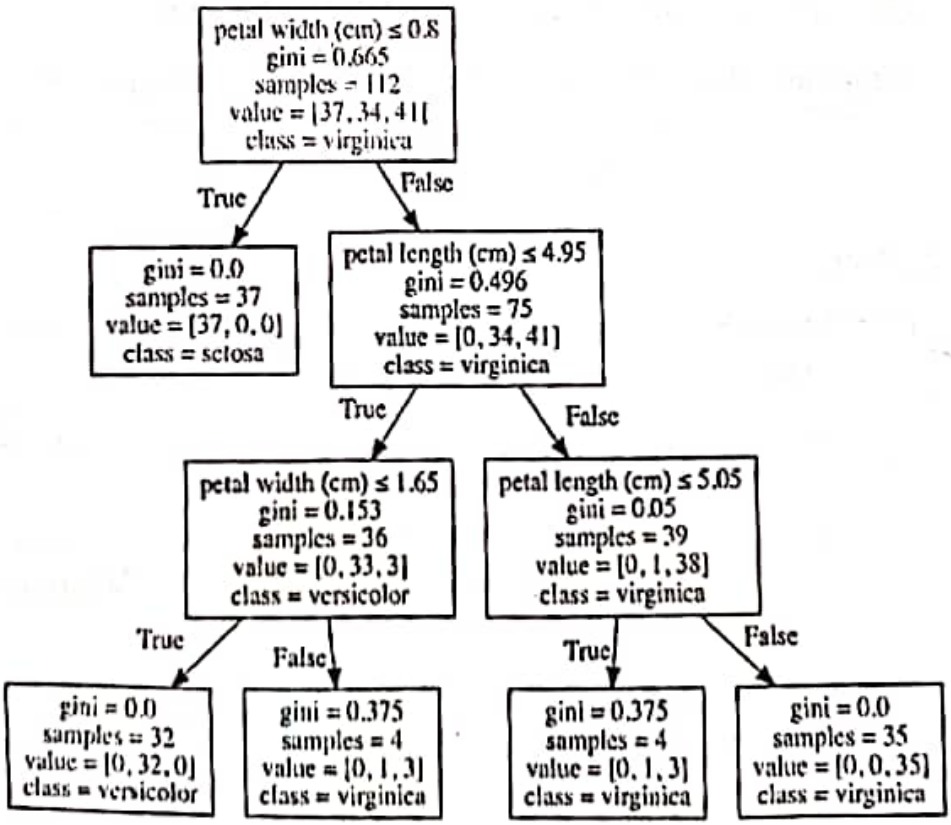
Analyze the following decision tree that classifies the flowers in iris dataset and answer the questions given below. Note that the class indexes [0, 1, 2] indicates class names [Setosa, Versicolor, Virginica].

(i) Mark the pure nodes in the tree.

(ii) Each node of the tree also shows winner class. How this is done ?

(iii) Which nodes in the tree can be further expanded ?

(iv) What is the accuracy of the model on training data ?



8(CO3)

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(b) (i) Explain the following SVM terminologies using appropriate graph of Linear classification with two features x1 and x2 :

(a) Support vectors.

(b) Maximum margin hyper plane.

(c) Margin.

(d) Positive and Negative hyper planes.

(e) Gamma and C paprameter.

(ii) Explain multiclass classification using one versus all and one versus one approaches with the help of neat schematics. Explain final classification step in both the cases. 8(CO3)

4. Solve any **One** :

(a) (i) What is agglomerative clustering ? Explain in brief about how dendrograms are useful in describing the output of hierarchical clustering.

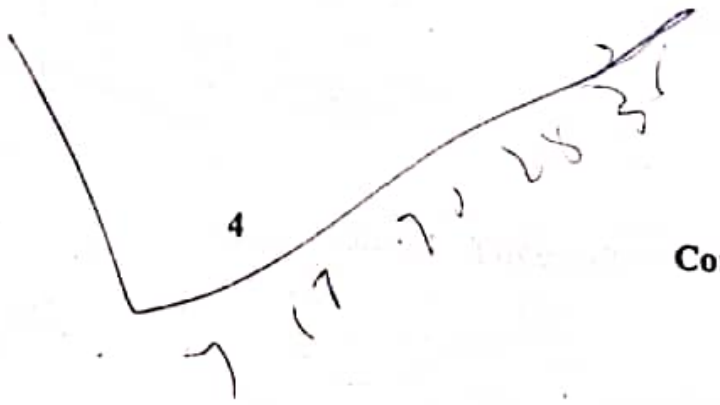
(ii) For the one dimensional data set {7,10,20,28,35}, hierarchical clustering is performed with the single linkage. What is the Euclidean distance for the highest level of the dendrogram ? 8(CO2)

(b) Write an algorithm to analyze top m outliers present in the dataset after it is clustered using a clustering algorithm, that gives k cluster centers as output. Assume that the dataset is unlabeled. 8(CO2)

5. Solve any **One** :

(a) (i) State the different methods for evaluating the performance of regression model.

(ii) Evaluate the performance of a regression model using Mean absolute error used to predict house prices. The model's predicted values are compared to the true values for a set of 10 houses. The **true values** (actual prices) and **predicted values** (model outputs) are given



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below :

House ID	True Price (in $000)	Predicted Price (in $000)
1	350	360
2	400	385
3	270	280
4	500	490
5	600	610
6	550	540
7	450	455
8	700	690
9	420	415
10	375	370

(iii) Explain what the MAE indicates about the model's performance. 9(CO4)

(b) (i) What is confusion matrix ?

(ii) Find the average accuracy and average F1 Score for the given matrix.

Predicted Class \ True Class	Apple	Orange	Mango
Apple	7	8	9
Orange	1	2	3
Mango	3	2	1

9(CO4)

6. Solve any **One** :

(a) (i) What is Softmax function ? Explain the softmax laver of CNN, with the help of suitable example.

(ii) Implement 3-input NAND gate using MP Neuron and show the diagram. 8(CO1)

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(b) Explain the design approach for gender classification based on the images of men and women using CNN. Also comment on the layers, activation functions of each layer and the design of image dataset to be used for training and testing. 8(CO1)

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1. Solve any **One** :

(a) Differentiate between traditional programming and machine learning. When evaluating or designing ML experiments, **data quality** plays a central role justify this statement in terms of Role of data, Characteristics of data and effect of noisy data of ML model. 8(CO1)

(b) (i) Differentiate between classification and regression.

(ii) A real - estate analyst trained a linear model :

$\hat{y} = 0.025x1 + 1.1x2 - 0.03x3 + 0.5$

Where the same feature meanings apply (x1 area in sq. ft., x2 bedrooms, x3 house age in years) and y is price in **lakhs**.
Find predicted price for a home with [x1, x2, x3] = [900, 2, 5]. Should this model be used for predicting prices of newly built apartments in a different neighborhood ? Briefly justify. 8(CO1)

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2. Solve any **One** :

(a) (i) Write the pseudocode for the KNN regression algorithm.

(ii) It is required to predict the sale price of used car based upon the age of a car and the mileage. The training data for the model is given below.

CarID	CarAge	Mileage	SalePrice
MH-31 AB 2333	2	11235	3.55
MH-40 EC 2445	5	25000	3.1
MH-40 AC 1901	2	120000	3.4
MH-40 BC 2435	2	25500	3.25
MH-29 ED 3495	4	30125	2.9
MH-34 AX 9999	6	45000	2.5
MH-31 AB 2333	5	20000	3.4
MH-31 DZ 5622	1	16000	3.95
MH-31 DC 4888	4	32000	3.5
MH-36 AD 5333	3	18904	3.65
MH-39 AX 9522	7	66580	2.1

Use KNN Regression model (with k = 2) to predict the sale price of used car with of age = 2 years and mileage = 23500 km. Apply min-max scalar and Euclidean distance. 8(CO2)

(b) (i) What is linear regression ?

(ii) The data from the early days of the Italian clothing company Benetton is given below.

Advertising : x = [23, 26, 30, 34, 43, 48, 52, 57, 58]

Sales : y = [651, 762, 856, 1063, 1190, 1298, 1421, 1440, 1518]

Each (x, y) shows Benetton's amount spent on advertising that year and the respective sales for that year. Apply linear regression to Predict sales for a given advertising amount of 45 million Euros. 8(CO2)

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3. Solve any **One** :

(a) (i) Write and explain the steps for ID3 algorithm.

(ii) Using the ID3 algorithm, calculate the entropy and information gain for each attribute and determine which should be chosen as the **root node** for building the decision tree.

ID	Age	Income	Student	CreditRating	BuysComputer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes

9(CO3)

(b) (i) Compare Decision Trees and Support Vector Machines (SVM) for classification problems.

(ii) For the One Vs. One classification using SVM, we have the following situation for a 4-class classifier, with class names as A, B, C, D with the feature vector $\mathbf{x}_{new}$:

SVM(A Vs. B) predicts A, SVM(A Vs. C) predicts A, SVM(D Vs. A) predicts A,

SVM(D Vs. B) predicts D, SVM(D Vs. C) predicts C, SVM(C Vs. B) predicts B.

Then what is the final classification for the feature vector $\mathbf{x}_{new}$? Justify your answer. 9(CO3)

Contd,

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4. Solve any **One** :

(a) (i) Differentiate between agglomerative and divisive clustering.

(ii) Consider the following 2D data points :
(2, 3)(3, 4)(5, 8)(8, 8)(1, 0)(9, 11)
Run one iteration of the k-means algorithm with k = 2 with initial centroid at (2, 3) and (5, 8).
Show cluster assignment for each point New centroids after first iteration. 9(CO2)

(b) What is a dendrogram ? Perform single link clustering on the following 2-D data with city block distance and draw the resulting dendrogram : [1, 3], [1, 4], [5, 2], [5, 1], [2, 2], [7, 2] 9(CO2)

5. Solve any **One** :

(a) (i) Write and explain strategies of experimentation in brief.

(ii) A data science team at a telecom company wants to develop a **machine learning model** to predict whether a customer will **churn** (leave the service) or **stay**. They have collected data containing customer demographics, usage details and billing information. The team plans to experiment with the following parameters :

Feature Selection Method Correlation-based / Mutual Information

Classifier Algorithm Decision Tree / Random Forest / Logistic Regression

Feature Scaling Method Min–Max Scaling / Standardization

Evaluation Metric Accuracy / F1-score

Which strategy of experimentation will give the **best model performance** ? Justify. 8(CO4)

(b) (i) What is confusion matrix and AUC ROC curve ?

(ii) A machine learning model is developed to classify whether a patient has a heart disease **(Positive)** or not **(Negative)** based on their medical data.

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When tested on 100 patients, the model predicted **40** patients as positive. Out of those 40 predicted positive, **30** actually had heart disease. Out of the 60 predicted negatives, **10** actually had heart disease.

(1) Construct the **confusion matrix** for this classification problem.

(2) Compute the **Accuracy**, **Precision**, **Recall** and **F1 Score** for the model. 8(CO4)

6. Solve any **One** :

(a) Explain three major layers of CNN with a schematic of CNN architecture. State application of the same. 8(CO1)

(b) (i) What is Softmax function ? Explain the softmax laver of CNN, with the help of suitable example.

(ii) Explain implementation of a memory cell using MP neuron, with the help of neat diagram. 8(CO1)

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Yeshwantrao Chavan College of Engineering, Nagpur
Department of Information Technology
ODD Term (Session 2025-26)
Mid Semester Examination-I
Subject: Machine Learning [23IT1504]

Semester: VI **Date:13.09.25**
Time: 1.5 Hrs. **Max. Marks: 30**

Solve any two sub questions from each question

Qu 1	a)	Differentiate between classification and regression. Consider a Credit Scoring application. The objective of credit scoring is to assign both existing and prospective customers to one of two groups: good or bad, depending on the applicant's demographics and credit bureau records and their recent financial transactions. A member of the good group is considered likely to repay their financial obligation like a loan. Is this a classification or regression problem? justify.	05	CO1	L2
	b)	What are the three types of Machine Learning algorithms? A publishing company wants to automatically organize a large collection of research articles into meaningful topics to help researchers easily browse articles related to their field of interest. They do not have any labeled data indicating the topic of each article. Which type of Machine Learning approach (Supervised or Unsupervised) is most suitable for this task and why? Give one commonly used algorithm that can be applied to this problem.	05	CO1	L2
	c)	A Multivariate linear model for Student's salary prediction for YCCE is trained and has the model coefficients [β_1 , β_2 , β_3] = [0.85, 0.86, 1.07] for [$CGPA, CommSkill, CodingSkill$] respectively and β_0 = -13.66. Then answer the following: • Is it a regression or classification Model? • Write the equation of the model. • If a student has $\boldsymbol{x} = [8.1, 8.0, 5.0]$, then what is his predicted salary ? • Can we use this model to predict the salary of a student of other colleges? Justify.	05	CO1	L2
Qu 2	a)	Consider the following table – it consists of the height, age and weight (target) value for 10 people. As you can see, the weight value of ID 11 is missing. Predict the weight of this person based on their height and age, using KNN with k=2 and the Manhattan Distance.	05	CO2	L3
	b)	Consider the dataset of some course students given in the following table. Use KNN algorithm to predict Final score for the student with study hours=11 and attendance=78%. Use the value of k = 3 and the Euclidean distance.	05	CO2	L3
	c)	What is feature independence assumption in Naïve Bayes classification? In a binary classification using Naive Bayes, a test instance has the following probabilities: P(Class A) = 0.4, P(Class B) = 0.6 P(x1|A) = 0.5, P(x2|A) = 0.2 P(x1|B) = 0.1, P(x2|B) = 0.8 Assuming feature independence, which class will the model assign to the instance with feature vector **x** = [x1, x2] ?	05	CO2	

ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	?

ID	Study Hours	Attendance	Final score
1	12	85	88
2	10	80	75
3	15	90	95
4	8	70	65
5	9	75	70
6	14	88	92
7	7	60	50
8	13	82	85

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Qu 3				
	a)	A decision tree regression model is trained to predict a person's annual medical expenses in INR based on two predictors: Age (years) and BMI (Body Mass Index). The dataset used for training is :	5	CO3

Age	BMI	Medical Expense
25	22.0	25000
35	28.0	40000
45	31.0	60000
50	29.0	55000
60	35.0	80000
30	25.0	30000
40	30.0	50000

- ☐ First split on **BMI** at 28.5:
  - If BMI ≤ 28.5 → Leaf Node A
  - If BMI > 28.5 → Next split on Age
- ☐ Next split on **Age** at 45:
  - If Age ≤ 45 → Leaf Node B
  - If Age > 45 → Leaf Node C

i) Draw the decision tree using these splits.
ii) Analyze the tree and find the predicted medical expense using mean for a patient aged **42 years** with **BMI = 29.0**

	b)	Consider the decision tree given below, that decides whether to play tennis or not in a given weather condition represented by feature vector [Temperature, Outlook, Wind, Humidity] and answer the following questions. It is given that there are 9 instances belonging to class YES (Play Tennis) and 5 instances belonging to the class NO (Do not play Tennis). Analyse the given tree and answer the following: i) How the node labelling (class labelling) takes place? ii) Redraw the tree with class labels written in place of "?" for each node. iii) Encircle the pure nodes in the tree. iv) What is the accuracy of the model on training data? v) What is the output of the decision tree if the test feature vector is [80, Overcast, Strong,85]		CO3

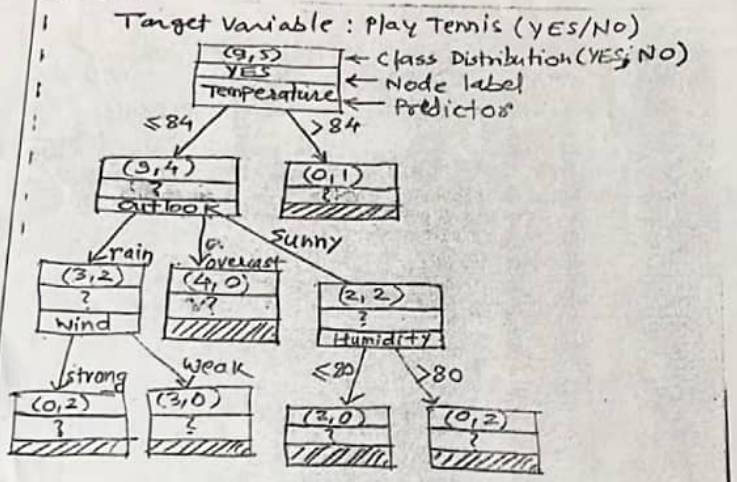
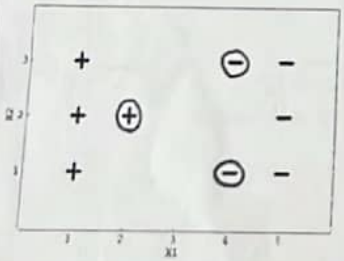


Fig. Decision Tree for playing Tennis & weather

	c)	Explain the following SVM terminologies using appropriate graph of Linear classification with two features x1 and x2. i) Support vectors ii) Maximum margin hyperplane iii) Margin iv) positive and negative hyperplanes Suppose you are using a Linear SVM classifier with binary classification problem. Now you have been given the following data in which some points are circled red that are representing support vectors. Then analyse the following vector space and answer the following question as True or False with Justification.	5	



i) If you remove the following any one red points from the data will the decision boundary change ?
ii) If you remove the non-red circled points from the data, will the decision boundary change?

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